IP addressing & Subnetting

1) List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

Class	IP Address Range	Default Subnet Mask	Real-World Example
A	1.0.0.0 – 126.255.255.255	255.0.0.0	Large organizations or Internet Service Providers (ISPs)
B	128.0.0.0 – 191.255.255.255	255.255.0.0	A university or large company network
C	192.0.0.0 – 223.255.255.255	255.255.255.0	Home Wi-Fi router or small business network
D	224.0.0.0 – 239.255.255.255	N/A	Multicast services such as live video streaming
E	240.0.0.0 – 255.255.255.255	N/A	Experimental and research purposes

2) Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Hint: Show your working steps for each calculation.

- Given:
  ip address 192.168,10.45
  subnet mask 255.255.255.0

  using subnet mask we can determine that ip belongs in **CIDR "/24"**
  so **network address** is **192.168.10.0**. There are total of 256 hosts in this network therefore we can determine that the **broadcast address** is **192.168.10.255**.
  since there are **256 hosts** the range of **valid host** addresses are
  192.168.10.1 to 192.168.10.254

3) Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

- Given : 10.0.0.0/24
  We are asked to divide network in 4 subnets
  We can equally divide it as follows 24/4 = 6.
  Therefore each subnet has 64 hosts and 62 valid host

  For Delivery Partners:
  - Network address: 10.0.0.0
  - Subnet mask:255.255.255.192
  - Broadcast : 10.0.0.63
  - Valid hosts range: 10.0.0.1 to 10.0.0.62

  for Restaurants:
  - Network address: 10.0.0.64
  - Subnet mask:255.255.255.
  - Broadcast : 10.0.0.127
  - Valid hosts range: 10.0.0.65 to 10.0.0.126

  for Customers:
  - Network address: 10.0.0.128
  - Subnet mask:255.255.255.
  - Broadcast : 10.0.0.191
  - Valid hosts range: 10.0.0.127 to 10.0.0.190

  for Admins:
  - Network address: 10.0.0.192
  - Subnet mask:255.255.255.
  - Broadcast : 10.0.0.255
  - Valid hosts range: 10.0.0.193 to 10.0.0.254

4) You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Constraint Show your calculation steps and final answers.

- **Given:**
- Network = **172.16.0.0/16**
- Need at least **20 subnets**

- 2^4 = 16 (Not enough)
- 2^5 = 32 (Enough)

- Borrow **5 bits**.

- Original = /16
- Borrowed = 5 bits

- New Prefix = /21

- 255.255.248.0

- 32 − 21 = 11 bits
- Hosts = 2^11 − 2
- = 2048 − 2
- = 2046 hosts

Final Answer

Item	**Value**
Original Network	**172.16.0.0/16**
Required Subnets	**20**
Borrowed Bits	**5**
New Prefix	**/21**
New Subnet Mask	**255.255.248.0**
Number of Subnets	**32**
Hosts per Subnet	**2046**

5) **A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.**

- The IP address 192.168.1.300 is invalid because each number (octet) in an IPv4 address must be between 0 and 255. The value 300 is outside this valid range, so the device cannot use that address.